Data and Algorithm Analysis

Chapter 5 — Probabilistic Analysis and Randomized Algorithms

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Outline

Introduction

Probability Theory

The Hiring Problem

Probabilistic Analysis

Randomized Algorithms

Table of Contents

Introduction

Probability Theory

The Hiring Problem

Probabilistic Analysis

Randomized Algorithms

Overview

- Understand the difference between probabilistic analysis and randomized algorithms.
- Indicator random variables technique.
- Analysis of a randomized algorithm.

Table of Contents

Introduction

Probability Theory

The Hiring Problem

Probabilistic Analysis

Randomized Algorithms

Probability Review

- Read Appendix C.2–C-4
- There is always a **sample space** S from which samples are drawn.
- Some of the subsets of S are **events**, namely, those to which a **probability distribution** $\Pr\{\}$ can be assigned.
- For every event $A \subseteq S$, we require $0 \leq \Pr\{A\} \leq 1$.

  Two events $A, B \subseteq$ are **mutually exclusive** if $A \cap B = \emptyset$.

Axioms of Probability

1. $\Pr\{A\} \geq 0$ for any event $A \subseteq S$.
2. $\Pr\{S\} = 1$.
3. $\Pr\{A \cup B\} = \Pr\{A\} + \Pr\{B\}$ for any two mutually exclusive events A and B.

   More generally, for any sequence of events $A_1, A_2, \ldots$ (finite or countably infinite) that are **pairwise mutually exclusive**.

$$\Pr\{\bigcup_i A_i\} = \sum_i \Pr\{A_i\}.$$

Conditional Probability

Given two events A and B, where $\Pr\{B\} > 0$, the **conditional probability of** A **given** B is the probability that A occurs, given that we know that B occurs.

The formula is simply

$$\Pr\{A \mid B\} = \frac{\Pr\{A \cap B\}}{\Pr\{B\}}.$$

Conditional probability shifts the sample space from S to B, making $\Pr\{B \mid B\} = 1$.

Independence

Two events A and B are **independent** if

$$\Pr\{A \cap B\} = \Pr\{A\}\Pr\{B\},$$

which is equivalent to

$$\Pr\{A \mid B\} = \Pr\{A\},$$

as long as $\Pr\{B\} > 0$.

A collection of events A_1, A_2 is **pairwise independent** if $\Pr\{A_i \cap A_j\} = \Pr\{A_i\}\Pr\{A_j\}$

Bayes' Theorem

Let A and B be events with nonzero probabilities. Using the definition of conditional independence twice, we can show that

$$\Pr\{A \mid B\} = \frac{\Pr\{A\}\Pr\{B \mid A\}}{\Pr\{B\}}.$$

This result is **Bayes' Theorem**, which is, in turn, the basis for **Bayesian statistics**.

Discrete Probability Distributions

If S is finite or countably infinite, then $\Pr\{\}$ is a **discrete probability distribution**.
Examples:

- **Uniform distribution** when S is finite.
- **Bernoulli distribution** with parameter p.
- **Binomial distribution** with parameters n and k.
- **Geometric distribution** with parameter p.

Discrete Random Variables

A function $X : S \to \mathbb{R}$ is a **random variable**; we will always consider only **discrete random variables**.

Example: For the toss of a pair of dice as the sample space, the function X that gives the sum of the individual die is a random variable.

- What values can X have?
- If the dice are fair, then what probability is associated with $X = 2$?

The event $X = x$ is $\{s \in S : X(s) = x\}$ and thus

$$\Pr\{X = x\} = \sum_{s \in S : X(s) = x} \Pr\{s\}$$

Expected Value

For a discrete random variable X, the **mean** or **expected value** of X is

$$\mathrm{E}\{X\} = \sum_{x} x \cdot \Pr\{X = x\}.$$

What is the expected value of X when X is the sum of two tossed dice?

Linearity of Expectation

Let $\{X_i\}$ be a countable set of random variables with corresponding constants a_i. Then, the expectation of a linear combination of the random variables equals the linear combination of the expectations of the individual random variables:

$$\mathrm{E}\left\{\sum_i a_i \cdot X_i\right\} = \sum_i a_i \cdot \mathrm{E}\{X_i\}.$$

Table of Contents

Introduction

Probability Theory

The Hiring Problem

Probabilistic Analysis

Randomized Algorithms

The Hiring Problem

- Employ an agency to hire a new assistant
- The agency sends one candidate per day for interview
- After interview, must immediately decide whether or not to hire that candidate
- If hired, fire the current assistant even if they are recently hired
- Interview cost: c_i / candidate (agency fee)
- Hiring cost: c_h / candidate (firing cost + agency fee)
- Assume that $c_h > c_i$

Goal: Determine what the price of this strategy will be

The Hiring Problem

Pseudocode HireAssistant(n):

```
curr = 0 // candidate 0 is a least-qualified dummy candidate
for i = 1 to n do
    interview candidate i
    if candidate i is better than curr
        curr = i
        hire candidate i
```

Cost: $O(nc_i + mc_h)$ (n candidates, m of whom are hired)

Worst Case Analysis

In the worst case, all n candidates are hired.

This happens if candidates appear in increasing order of quality.

The cost of hiring all is $O(c_i n + c_h n) = O(c_h n)$

Table of Contents

Introduction

Probability Theory

The Hiring Problem

Probabilistic Analysis

Randomized Algorithms

Probabilistic Analysis

We have no control over the order in which candidates appear.

But we could assume that they come in a **random** order:

- Assign a rank to each candidate: rank(i): unique int in $[1, \ldots, n]$.
- $\langle \text{rank}(1), \ldots, \text{rank}(n) \rangle$ is a permutation of $\langle 1, \ldots, n \rangle$.
- The list of ranks is one of the $n!$ permutations.
- Each of the possible $n!$ permutations appears with equal probability (**uniform random permutation**).

Use knowledge of (or make assumptions) the distribution of inputs.

- The expectation is over this distribution
- Require to make a reasonable characterization of the input distribution

Randomized Algorithm

Might not know the distribution of inputs, or might not be able to model it computationally.

Instead, use randomization within the algorithm in order to impose a distribution on the inputs.

Modify the hiring problem scenario:

- Ask the agency sends the list of n candidates in advance.
- On each day, randomly choose a candidate to interview.
- Take control of the process and enforce a random order!

Randomized Algorithm

Definition (Randomized Algorithm)

An algorithm is *randomized* if its behavior is determined in part by values produced by a **random-number generator**.

RANDOM(a, b): returns an integer r, where $a \leq r \leq b$ and each of the possible values of r is equally likely.

In practice, RANDOM is implemented by a **pseudorandom-number generator**.

Indicator Random Variables

Given a sample space and an event A, an indicator random variable is defined as:

$$I\{A\} = \left\{ \begin{array}{ll} 1 & \text{if } A \text{ occur,} \\ 0 & \text{if } A \text{ does not occur} \end{array} \right\}. \tag{1}$$

Lemma

Given a sample space S and an event A in the sample space S, let $X_A = I\{A\}$. Then $E[X_A] = \Pr\{A\}$

Proof

Indicator Random Variables

Example

Determine the expected number of heads from one flip of a fair coin

- Sample space is $\{H, T\}$
- $\Pr\{H\} = \Pr\{T\} = 1/2$
- Define indicator random variable $X_H = I\{H\}$, i.e., X_H counts the number of heads in one flip.
- Since $Pr\{H\} = 1/2$, $E[X_H] = 1/2$ (due to above Lemma)

Indicator Random Variables

Example

Determine the expected number of heads in n flips

Analysis of the Hiring Problem

Assume that the candidates arrive in a random order.

Let X be a random variable indicating the number of times a new office assistant is hired.

Define indicator random variables $X_1, \ldots, X_n$ s.t.

$$X_i = I\{\text{candidate } i \text{ is hired}\}.$$

Properties:

- $X = X_1 + X_2 + \cdots + X_n$
- $E[X_i] = \Pr\{\text{candidate } i \text{ is hired}\}$

Goal: Compute $\Pr\{\text{candidate } i \text{ is hired}\}$

Analysis of the Hiring Problem

- Candidate i is hired iff candidate i is better than candidates $1, 2, \ldots, i-1$.
- Assumption candidates arrive in random order $\Rightarrow$ candidates $1, 2, \ldots, i$ arrive in random order $\Rightarrow$ any one of first i candidates is equally likely to be the best one so far.
  - $\Pr\{\text{candidate } i \text{ is the best so far}\} = 1/i \Rightarrow E[X_i] = 1/i$

$$\begin{aligned} E[X] = E\left[\sum_{i=1}^{n} X_i\right] &= \sum_{i=1}^{n} E[X_i] \\ = \sum_{i=1}^{n} 1/i &= \ln n + O(1) \end{aligned} \tag{2}$$

Expected hiring cost is $O(c_h \ln n)$.

Table of Contents

Introduction

Probability Theory

The Hiring Problem

Probabilistic Analysis

Randomized Algorithms

Randomized Algorithms

Instead of assuming a distribution of inputs, impose a distribution.

For the hiring problem, the algorithm is **deterministic**

- For any given input, the number of times of hiring a new assistant will always be the same.
- The number of times of hiring an assistant depends only on the input.
- In fact, it depends only on the ordering of the candidates' ranks that it is given.
  - Some rank orderings produces high hiring cost, e.g., $\langle 1, 2, 3, 4 \rangle$.
  - Some produces low hiring cost,e.g., best candidates interview first.
  - Some in between.

Randomized Algorithms

Instead of interviewing candidates in the defined order, randomly permute this order.

Randomization happens in **algorithm level**, not in the input distribution.

Can no longer say what its hiring cost will be given a particular input.

- Each run of the algorithm can result in a different hiring cost.
- Execution depends on the random choices made.
- No particular input always elicits worst-case behavior.
- Bad behavior occurs only if RNG generates “unlucky” numbers.

Randomized Hiring Problem

Pseudocode

RandomHireAssistant(n):

```
randomly permute the list of candidates
HireAssistant(n)
```

Expected hiring cost is $O(c_h \ln n)$.

Randomly Permute an Array

RandomlyPermute(A, n):

for $i = 1$ to n do
  swap $A[i]$ with $A[\text{Random}(i, n)]$

Complexity: $O(n)$

Correctness: Given a set of n elements, a k-permutation is a sequence containing k of the n elements. There are $n!/(n-k)!$ possible k-permutation.

Lemma

RandomlyPermute *computes a uniform random distribution.*